

# **GEOGRAPHY**

# **INDIAN GEOGRAPHY**



## **INDIAN CLIMATE**

### **MODULE - 7**

### **INDIAN MONSOON**

- **RAINFALL DISTRIBUTION & VARIABILITY**

## TRADITIONAL INDIAN SEASONS

| Seasons  | Months<br>(According to the<br>Indian Calendar) | Months<br>(According to the<br>Indian Calendar) |
|----------|-------------------------------------------------|-------------------------------------------------|
| Vasanta  | Chaitra-Vaisakha                                | March-April                                     |
| Grishma  | Jyaistha-Asadha                                 | May-June                                        |
| Varsha   | Sravana-Bhadra                                  | July-August                                     |
| Sharada  | Asvina-Kartika                                  | September-<br>October                           |
| Hemanta  | Margashirsa-<br>Pausa                           | November-<br>December                           |
| Shishira | Magha-Phalguna                                  | January-<br>February                            |

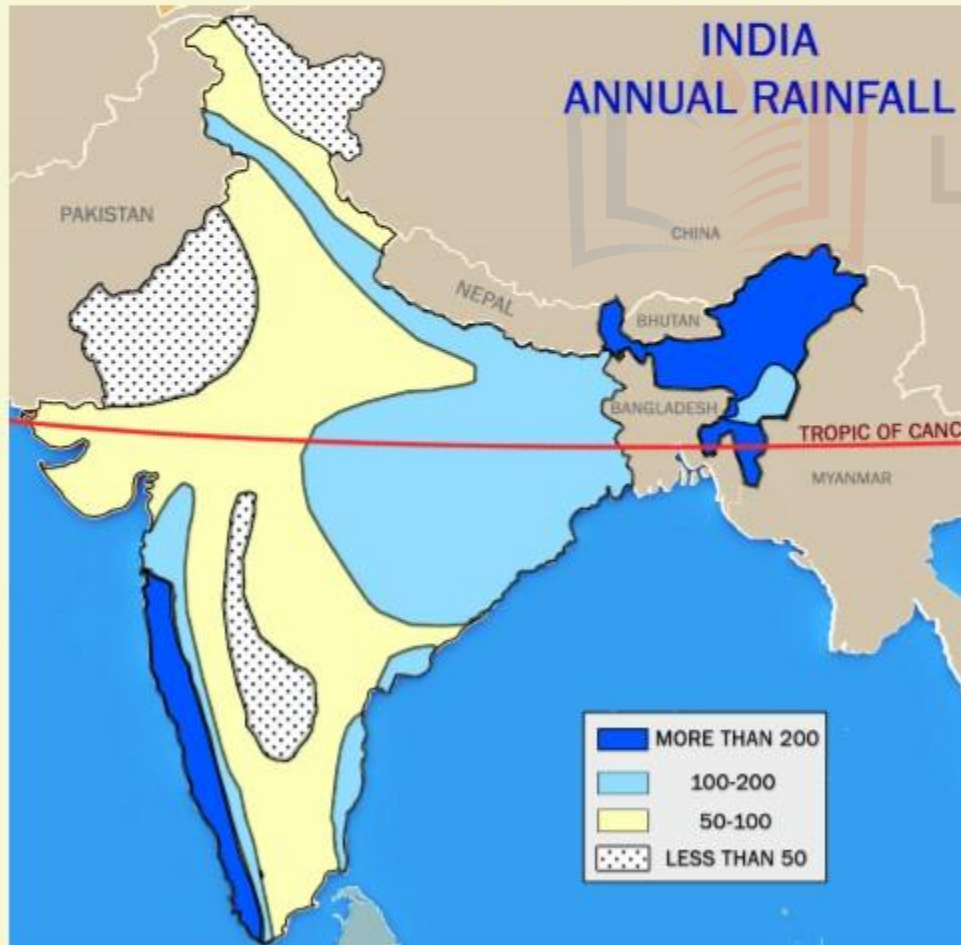
- 1 In the Indian tradition, a year is divided into six two-monthly seasons.
- 2 This cycle of seasons, which the common people in north and central India follow is based on their practical experience and age-old perception of weather phenomena.
- 3 However, this system does not match with the seasons of south India where there is little variation in the seasons.



# DISTRIBUTION OF RAINFALL

The average annual rainfall in India is about 125 cm, but it has great spatial variations

## 1 AREAS OF HIGH RAINFALL



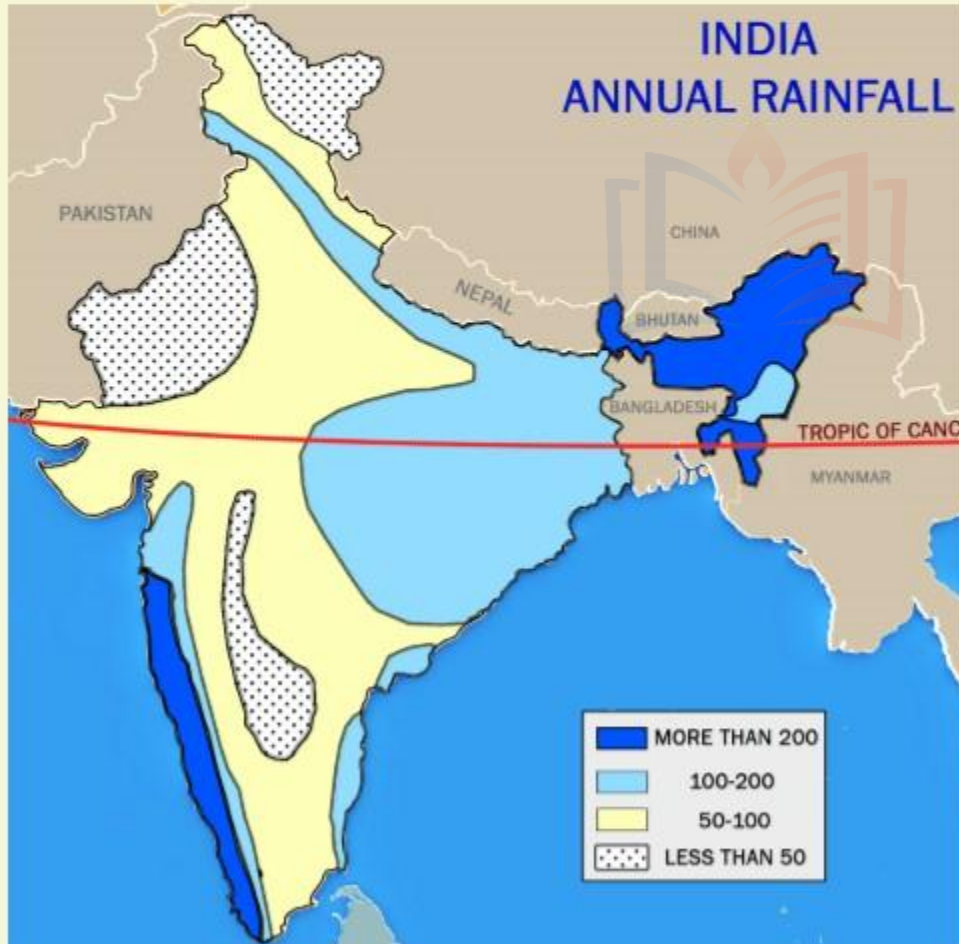
**1** The highest rainfall occurs along the west coast on the Western Ghats, as well as in the sub-Himalayan areas- the northeast and the hills of Meghalaya. Here the rainfall exceeds 200 cm.

**2** In some parts of Khasi and Jaintia hills, the rainfall exceeds 1,000 cm.

**3** In the Brahmaputra valley and the adjoining hills, the rainfall is less than 200 cm.

# DISTRIBUTION OF RAINFALL

## 2 AREAS OF MEDIUM RAINFALL

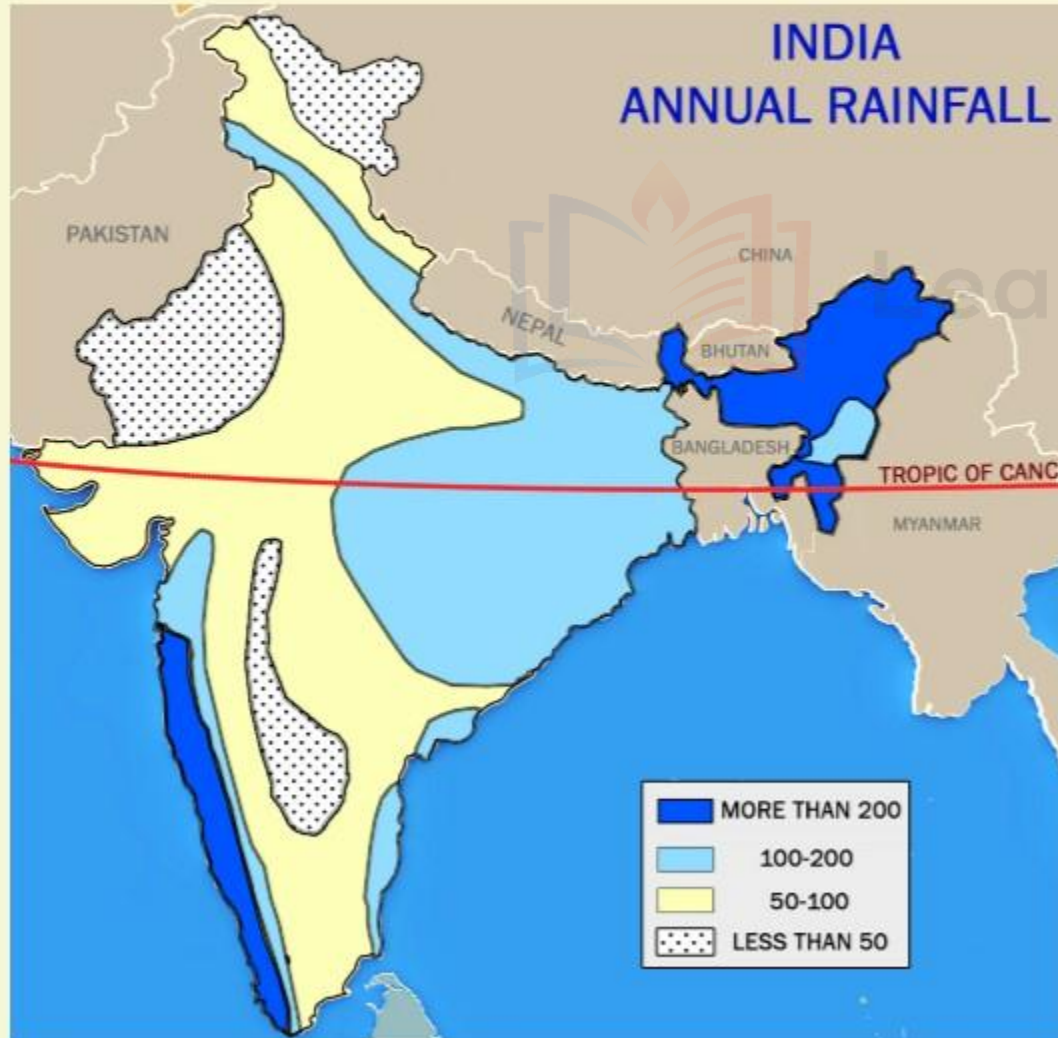


- 1 Rainfall between 100-200 cm is received in the southern parts of Gujarat, east Tamil Nadu, northeastern Peninsula covering Orissa, Jharkhand, Bihar, eastern Madhya Pradesh, northern Ganga plain along the sub-Himalayas and the Cachar Valley and Manipur.



# DISTRIBUTION OF RAINFALL

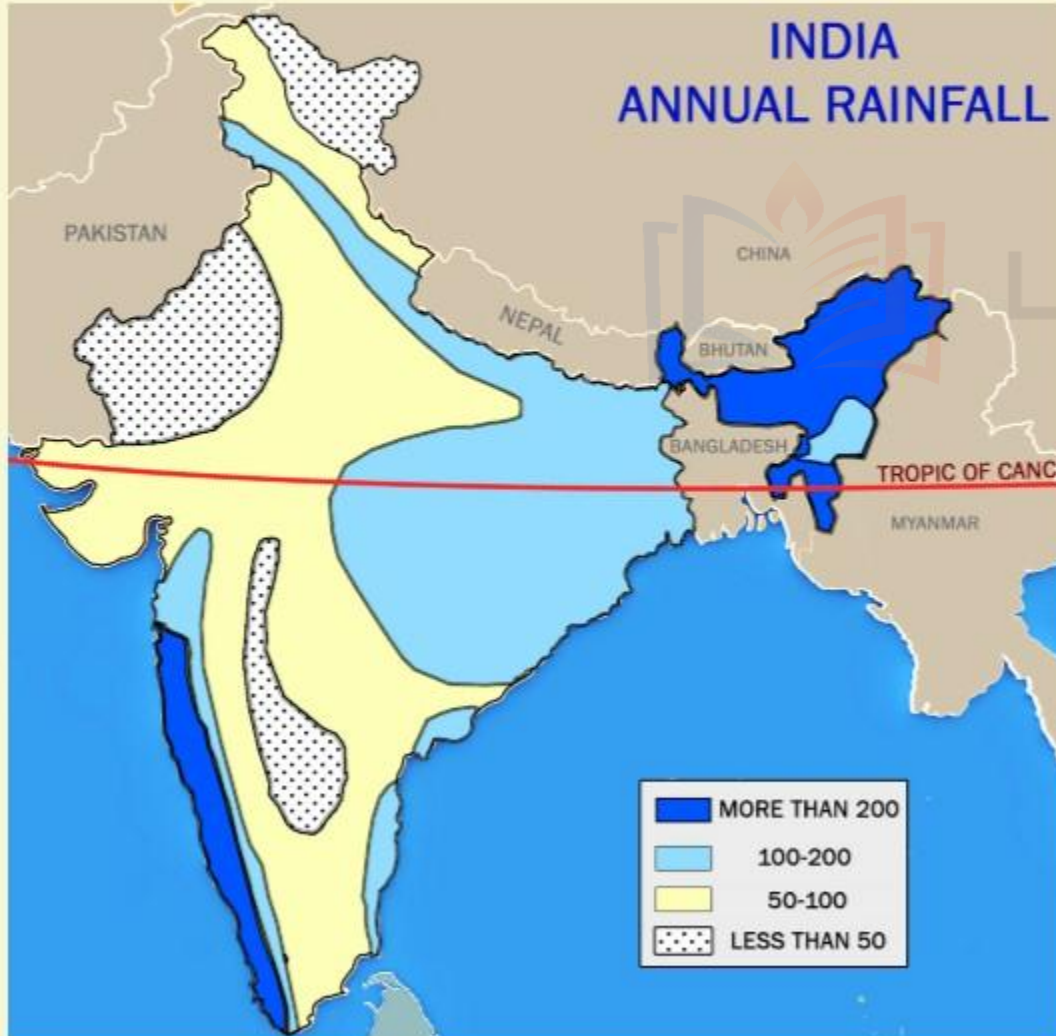
## 3 AREAS OF LOW RAINFALL



- 1 Western Uttar Pradesh, Delhi, Haryana, Punjab, Jammu and Kashmir, eastern Rajasthan, Gujarat and Deccan Plateau receive rainfall between 50-100 cm.

# DISTRIBUTION OF RAINFALL

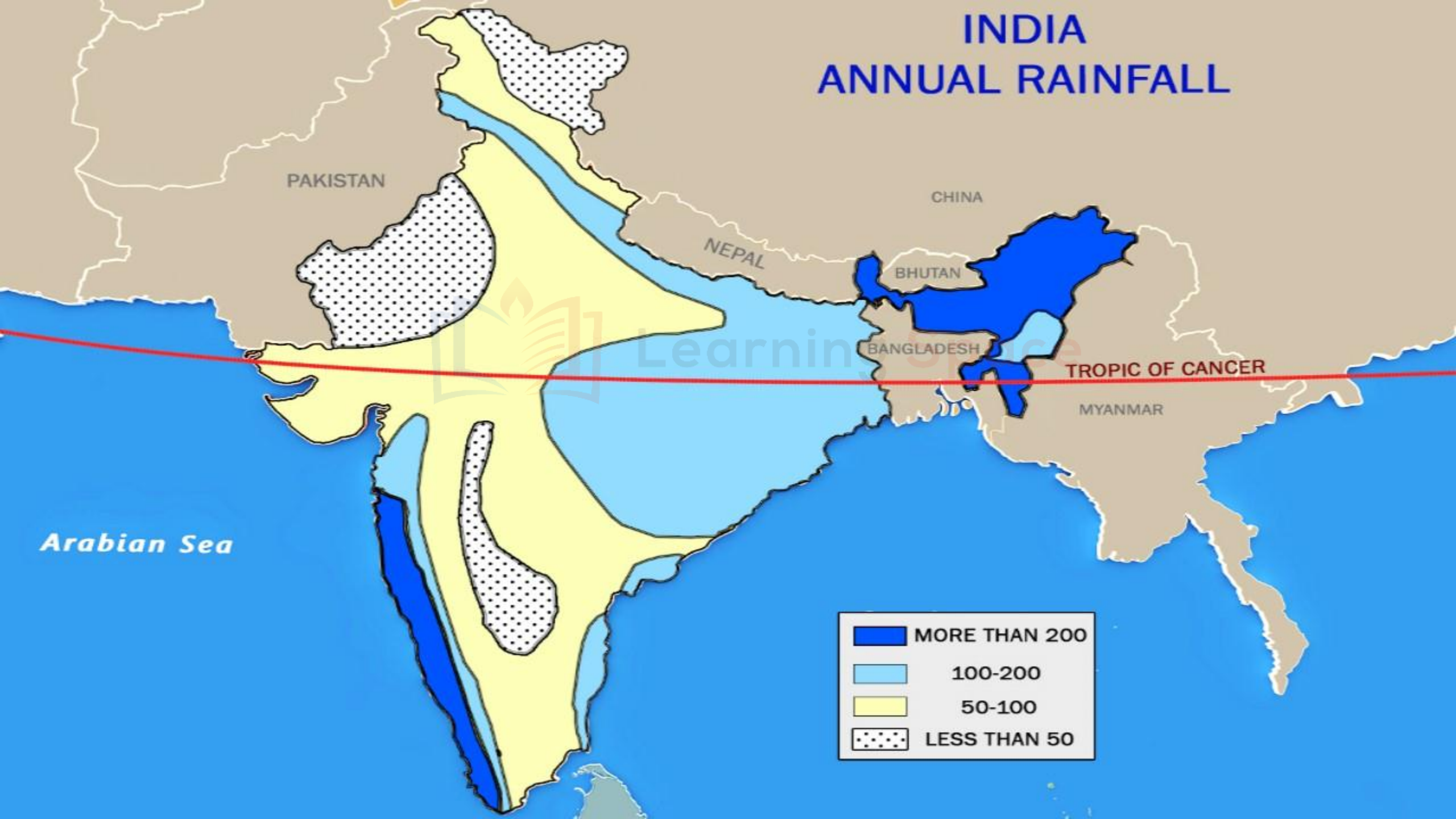
## 4 AREAS OF INADEQUATE RAINFALL



- 1** Parts of the Peninsula, especially in Andhra Pradesh, Karnataka and Maharashtra, Ladakh and most of western Rajasthan receive rainfall below 50 cm.
- 2** Snowfall is restricted to the Himalayan region.

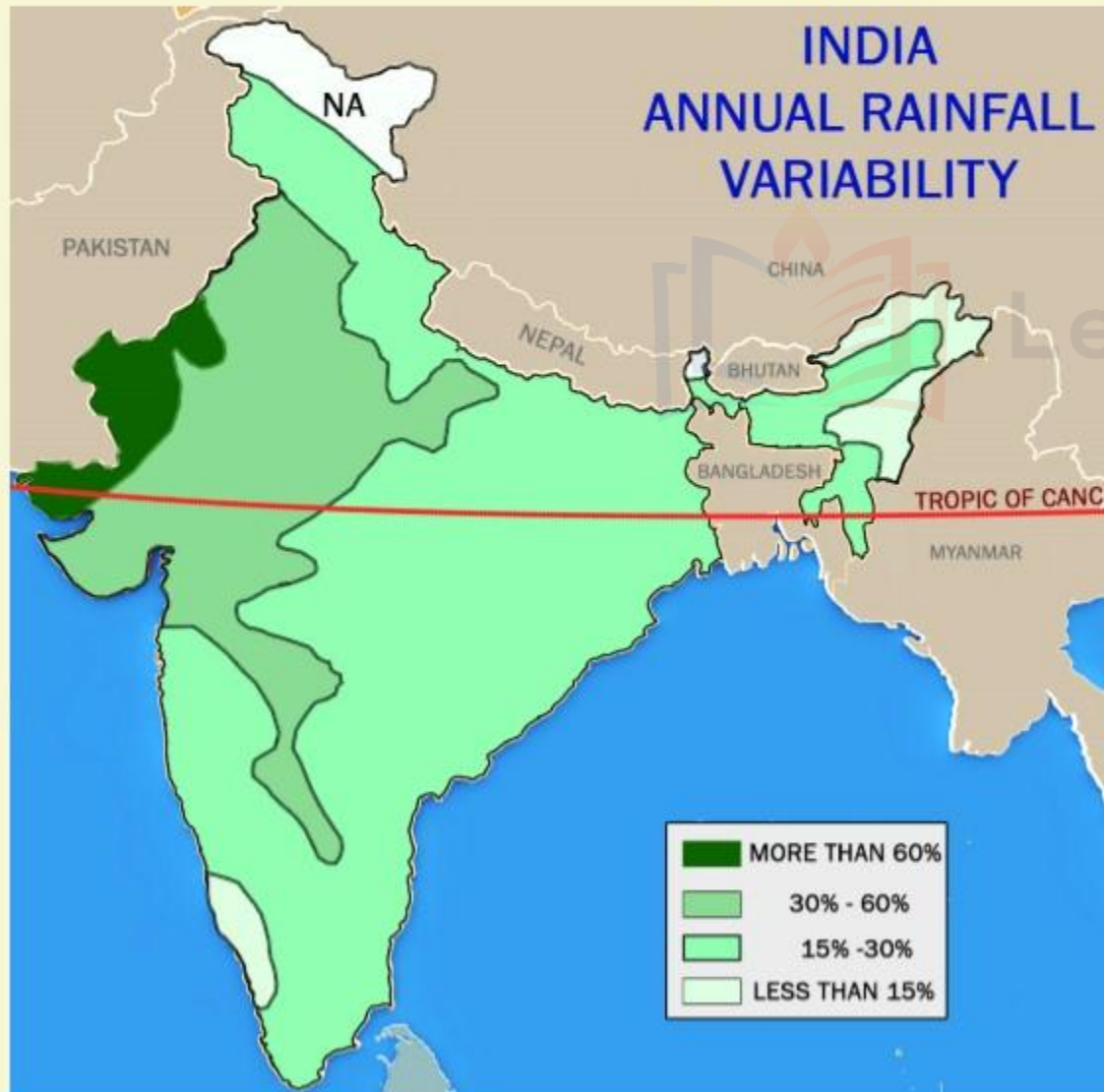


# INDIA ANNUAL RAINFALL



# DISTRIBUTION OF RAINFALL

## VARIABILITY OF RAINFALL



- 1 A characteristic feature of rainfall in India is its variability. The variability of rainfall is computed with the help of the following formula:

$$C.V. = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

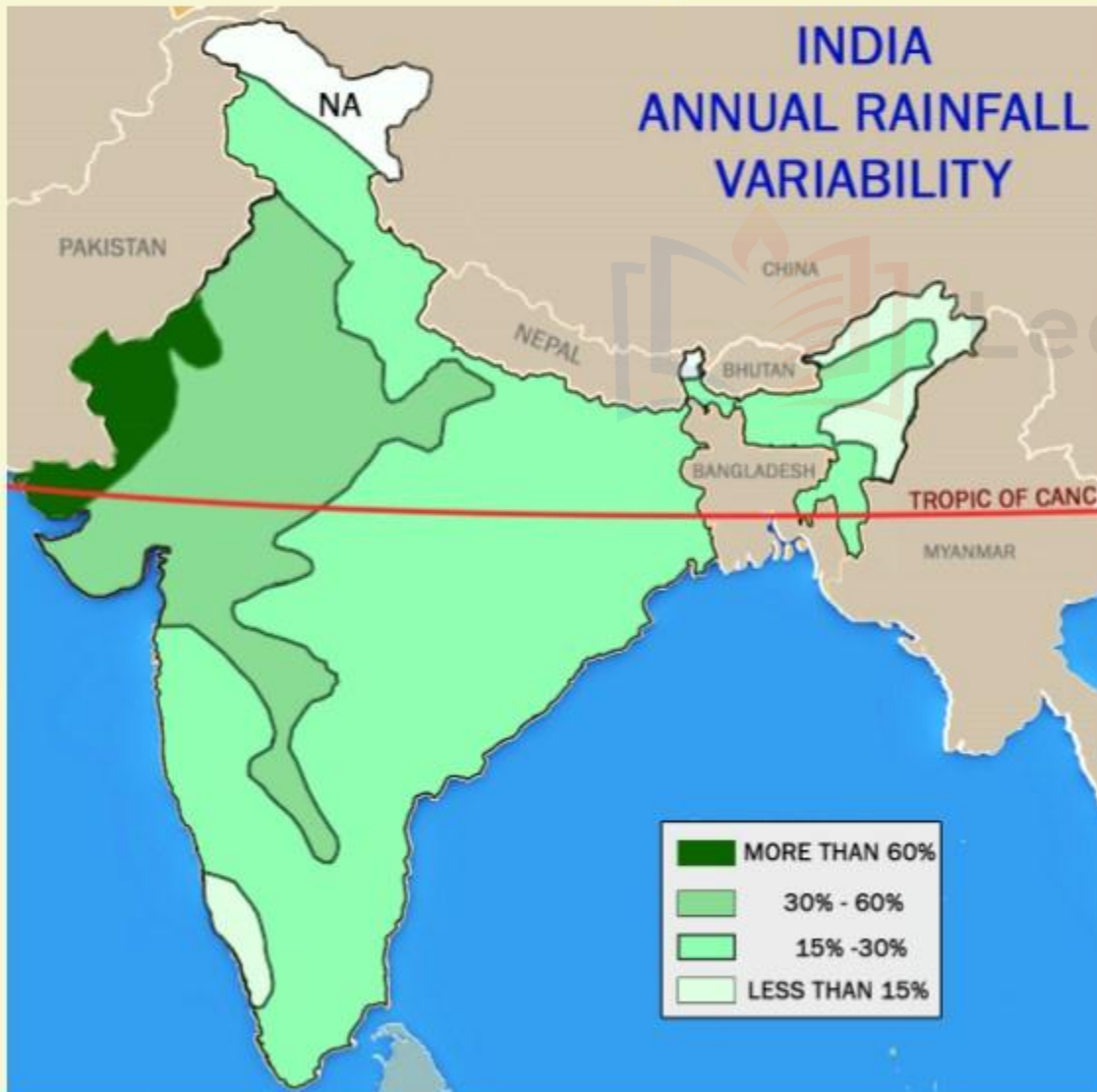
where C.V. is the coefficient of variation.

- 2 The values of coefficient of variation show the change from the mean values of rainfall.
- 3 The actual rainfall in some places deviates from 20-50 per cent.
- 4 The values of the co-efficient of variation show variability of rainfall in India.



# DISTRIBUTION OF RAINFALL

## VARIABILITY OF RAINFALL

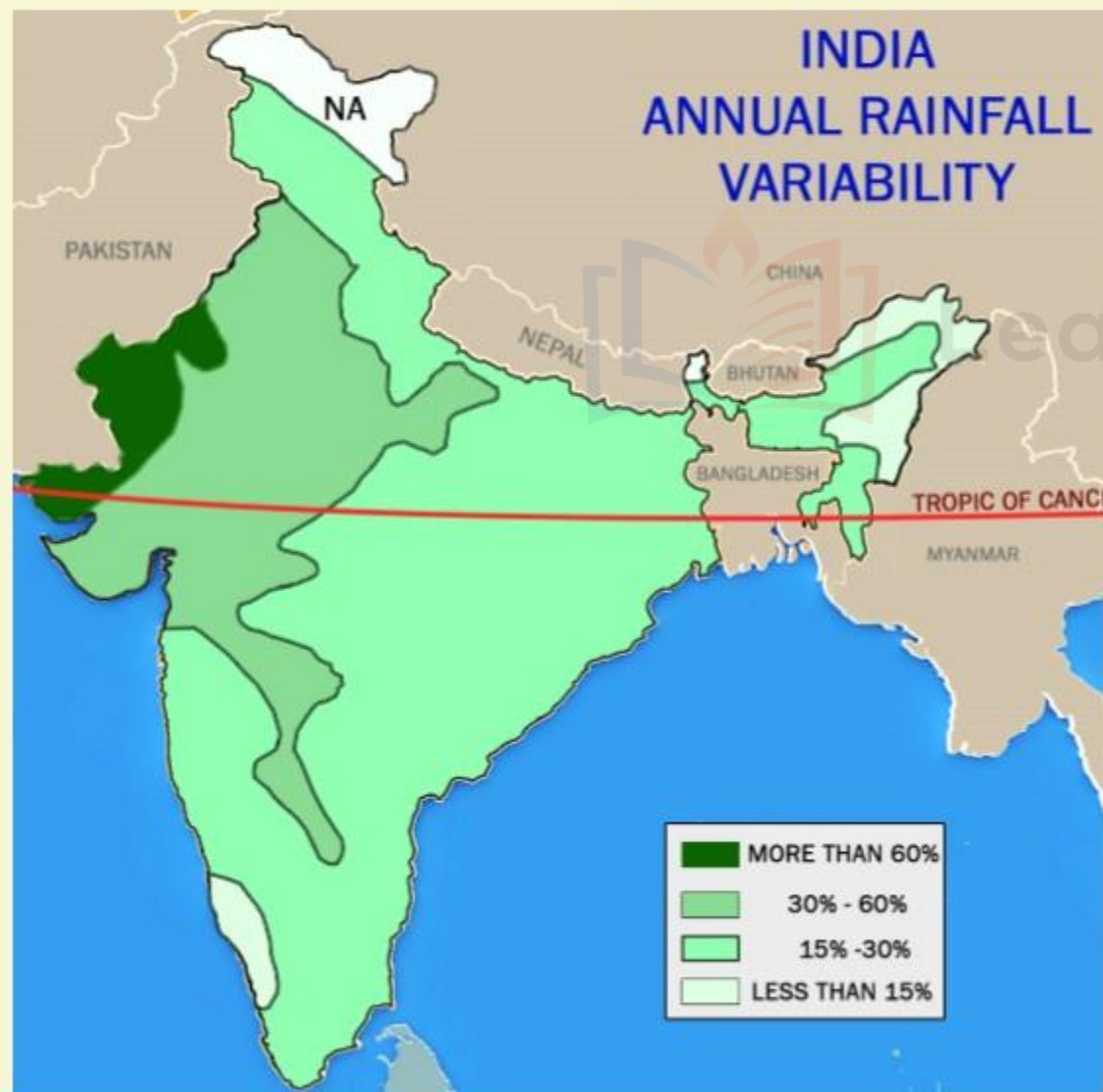


5

A variability of less than 25 per cent exists on the western coasts, Western Ghats, North-Eastern Peninsula, eastern plains of the Ganga, North-Eastern India, Uttaranchal and Himachal Pradesh and south-western part of Jammu and Kashmir. These areas have an annual rainfall of over 100 cm.

# DISTRIBUTION OF RAINFALL

## VARIABILITY OF RAINFALL

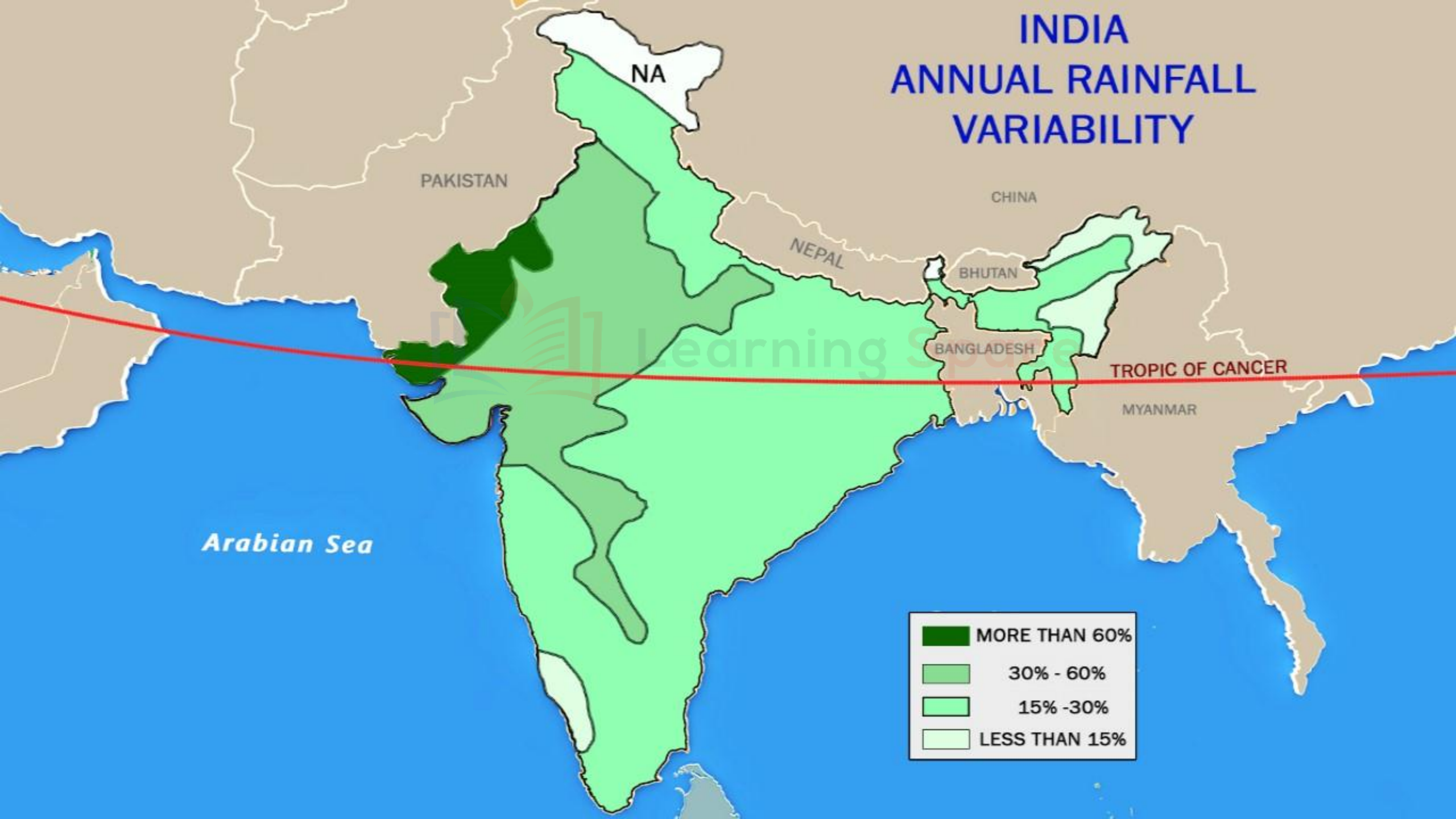


**6** A variability of over 50 per cent exists in the western part of Rajasthan and interior parts of the Deccan plateau. These areas have an annual rainfall of less than 50 cm.

**7** Rest of India have a variability of 25-50 per cent and these areas receive an annual rainfall between 50 -100 cm.



# INDIA ANNUAL RAINFALL VARIABILITY



## CLIMATIC REGIONS OF INDIA

- 1 The whole of India has a monsoon type of climate.
- 2 But the combination of elements of the weather, however, reveal many regional variations.
- 3 These variations represent the sub-types of the monsoon climate, it is on this basis that the climatic regions can be identified.
- 4 A climatic region has a homogeneous climatic condition which is the result of a combination of factors.
- 5 Temperature and rainfall are two important elements which are considered to be decisive in all the schemes of climatic classification.
- 6 The classification of climate, however, is a complex exercise. There are different schemes of classification of climate.
- 8 Major climatic types of India based on Koeppen's scheme have been described below.
- 9 Koeppen based his scheme of climatic classification on monthly values of temperature and precipitation.



## CLIMATIC REGIONS OF INDIA

- 10** He identified five major climatic types, namely:
- I. **Tropical climates (A)**, where mean monthly temperature throughout the year is over  $18^{\circ}\text{C}$ .
  - II. **Dry climates (B)**, where precipitation is very low in comparison to temperature, and hence, dry. If dryness is less, it is semiarid(S); if it is more, the climate is arid(W).
  - III. **Warm temperate climates (C)**, where mean temperature of the coldest month is between  $18^{\circ}\text{C}$  and minus  $3^{\circ}\text{C}$ .
  - IV. **Cool temperate climates (D)**, where mean temperature of the warmest month is over  $10^{\circ}\text{C}$ , and mean temperature of the coldest month is under minus  $3^{\circ}\text{C}$ .
  - V. **Ice climates (E)**, where mean temperature of the warmest month is under  $10^{\circ}\text{C}$ .

# CLIMATIC REGIONS OF INDIA



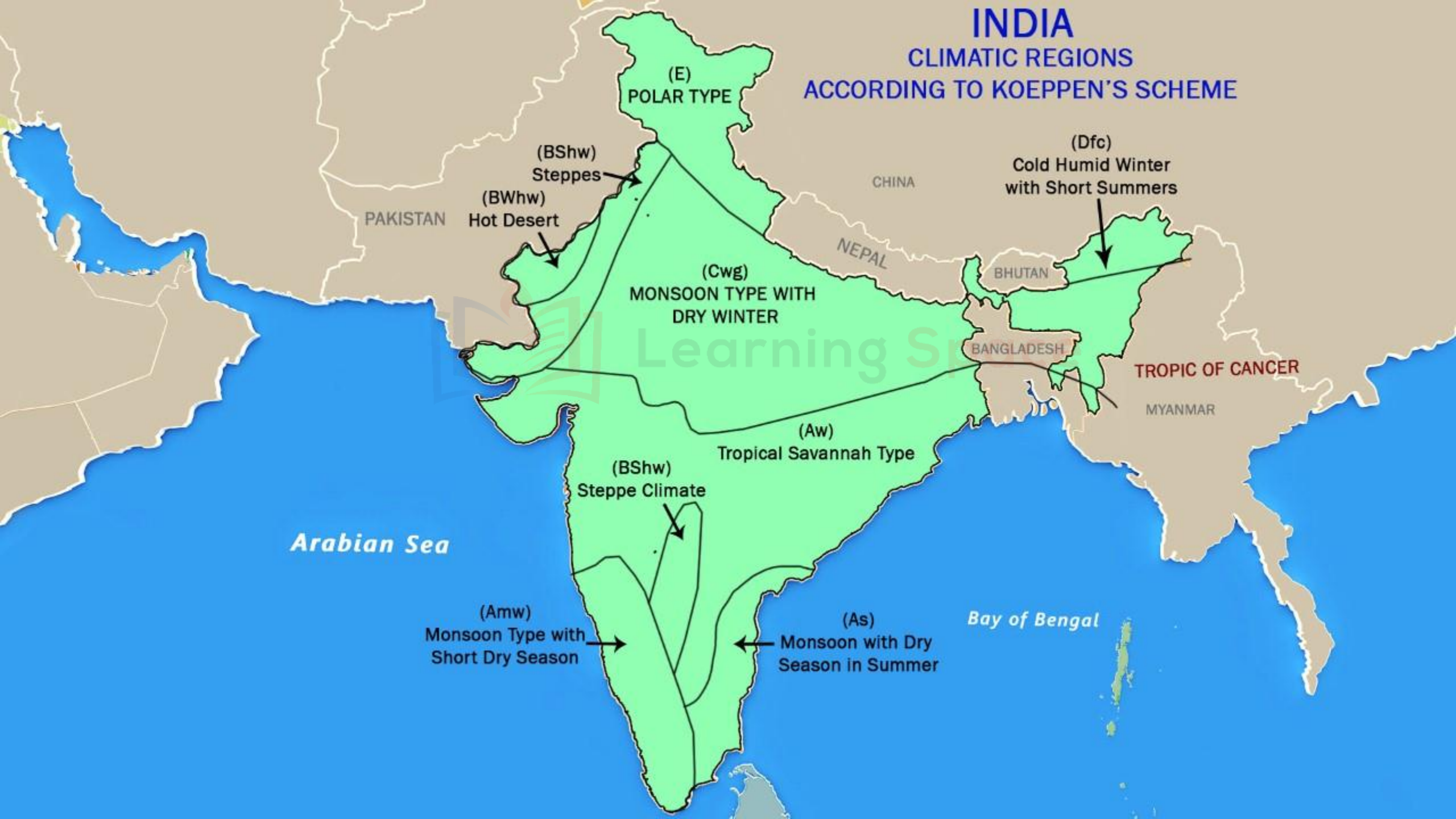
**11** Köppen used letter symbols to denote climatic types as given above. Each type is further sub-divided into sub-types on the basis of seasonal variations in the distributional pattern of rainfall and temperature.

**12** He used **S** for semi-arid and **W** for arid and the following small letters to define sub-types: **f** (sufficient precipitation), **m** (rainforest despite a dry monsoon season), **w** (dry season in winter), **h** (dry and hot), **c** (less than four months with mean temperature over  $10^{\circ}\text{C}$ ), and **g** (Gangetic plain).



# INDIA

## CLIMATIC REGIONS ACCORDING TO KOEPPEN'S SCHEME



## CLIMATIC REGIONS OF INDIA

*Accordingly, India can be divided into eight climatic regions:*

| TYPE OF CLIMATE                           | AREAS                                                                             |
|-------------------------------------------|-----------------------------------------------------------------------------------|
| Amw- Monsoon with short dry season        | West coast of India south of Goa                                                  |
| As – Monsoon with dry summer              | Coromandel coast of Tamil Nadu                                                    |
| Aw – Tropical savannah                    | Most of the Peninsular plateaus, south of the Tropic of Cancer                    |
| BShw – Semi-arid steppe climate           | North-western Gujarat, some parts of western Rajasthan and Punjab                 |
| BWhw – Hot desert                         | Extreme western Rajasthan                                                         |
| Cwg – Monsoon with dry winter             | Ganga plain, eastern Rajasthan, northern Madhya Pradesh, most of North-east India |
| Dfc – Cold humid winter with short summer | Arunachal Pradesh                                                                 |
| E – Polar type                            | Jammu and Kashmir, Himachal Pradesh and Uttarakhand                               |



# MONSOONS AND THE ECONOMIC LIFE IN INDIA



**1** Monsoon is that axis around which revolves the entire agricultural cycle of India. It is because about 64 per cent people of India depend on agriculture for their livelihood and agriculture itself is based on southwest monsoon.

**2** Except Himalayas all the parts of the country have temperature above the threshold level to grow the crops or plants throughout the year.

**3** Regional variations in monsoon climate help in growing various types of crops.

**4** Variability of rainfall brings droughts or floods every year in some parts of the country.



# MONSOONS AND THE ECONOMIC LIFE IN INDIA



**5** Agricultural prosperity of India depends very much on timely and adequately distributed rainfall. If it fails, agriculture is adversely affected particularly in those regions where means of irrigation are not developed.

**6** Sudden monsoon burst creates problem of soil erosion over large areas in India.

**7** Winter rainfall by temperate cyclones in north India is highly beneficial for rabi crops.

**8** Regional climatic variation in India is reflected in the vast variety of food, clothes and house types.



WE EXPRESS OUR  
SINCERE THANKS  
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

[suggestions@learningspace.in](mailto:suggestions@learningspace.in)

To Contact us:

[info@learningspace.in](mailto:info@learningspace.in)

**OUR TEAM**

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

[www.learningspacedigital.com](http://www.learningspacedigital.com)

0 8 6 6 -2 4 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9